A First Course in Abstract Algebra

Section 52: Totally Inseparable Extensions

Detailed worked solutions

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Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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52 Totally Inseparable Extensions

For degree checks involving two independent indeterminates, one useful elementary rule is

$$[\mathbb{F}_3(y,z) : \mathbb{F}_3(y^a z^b, y^c z^d)] = |ad - bc|$$

when the two exponent vectors are independent. Here is why the determinant appears. The subgroup of $\mathbb{Z}^2$ generated by $(a,b),(c,d)$ has $|ad-bc|$ cosets. Choose one monomial for each exponent coset. Every Laurent monomial is a coefficient-field monomial times one chosen representative, so these representatives span; their span is a finite-dimensional domain over the coefficient field, hence is a field and includes rational functions too. They are independent because clearing coefficient denominators leaves terms in distinct exponent cosets, which cannot cancel. Thus they form a basis. This supplies the degree certificates below; separability and the power tests establish the actual closure fields.

Exercise 52.1: A separable closure with two independent exponents

Let $K = \mathbb{Z}_3(y^{12}, z^{18})$ and $L = \mathbb{Z}_3(y,z)$. Set $S = \mathbb{Z}_3(y^3, z^9)$. The generators y^3 and z^9 satisfy respectively $X^4 - y^{12}$ and $X^2 - z^{18}$ over K, whose nonzero roots are simple in characteristic 3. Thus S/K is separable. Also $y^3, z^9 \in S$, so every element of L has some 3-power in S; in fact its ninth power is in S. Hence L/S is purely inseparable. An element separable over K is separable over S too, since its new minimal polynomial divides its old separable one; if it also lies in the purely inseparable extension L/S, it must be in S. Therefore S is exactly the separable closure. The monomial degree counts are $[L:K] = 12 \cdot 18 = 216$, $[L:S] = 3 \cdot 9 = 27$, and $[S:K] = 8$.

Exercise 52.2: A separable closure with a mixed monomial

Now $K = \mathbb{Z}_3(y^{12}, y^2 z^{18})$. Put $S = \mathbb{Z}_3(y^3, yz^9)$. The first generator satisfies the separable equation $X^4 - y^{12} = 0$ and the second the separable equation $X^2 - y^2 z^{18} = 0$, so S/K is separable. Moreover $y^3 \in S$ and $z^{27} = (yz^9)^3/y^3 \in S$. Thus $L = \mathbb{Z}_3(y,z)$ is purely inseparable over S. The separable/purely-inseparable intersection argument from Exercise 52.1 makes S the full separable closure. For a degree check, the exponent lattice of K has determinant $12 \cdot 18 = 216$, while that of S, generated by $(3,0),(1,9)$, has determinant 27; hence $[S:K] = 8$.

Exercise 52.3: A totally inseparable closure in the independent case

The field is $P = \mathbb{Z}_3(y^4, z^2)$. Its generators have a third and a ninth power, respectively, in $K = \mathbb{Z}_3(y^{12}, z^{18})$, so P/K is purely inseparable. The extension L/P is separable, because y satisfies $X^4 - y^4$ and z satisfies $X^2 - z^2$, both separable polynomials. If $a \in L$ has a 3-power in K, it is purely inseparable over P but also separable over P, so $a \in P$. This proves maximality. Thus the totally inseparable closure is P; its degree over K is 27.

Exercise 52.4: A totally inseparable closure in the mixed case

Take $P = \mathbb{Z}_3(y^4, y^2 z^2)$. It contains $K = \mathbb{Z}_3(y^{12}, y^2 z^{18})$, since $y^{12} = (y^4)^3$ and $y^2 z^{18} = (y^2 z^2)^9/(y^4)^4$. Its generators have 3-powers in K:

$$(y^4)^3 = y^{12}, \qquad (y^2 z^2)^{27} = (y^{12})^4 (y^2 z^{18})^3.$$

Hence P/K is purely inseparable. Over P, first adjoining y is separable via its fourth-power equation, and then adjoining z is separable via $z^2 = (y^2z^2)/y^2$. Thus L/P is separable. The same intersection argument as in Exercise 52.3 shows that every element of L purely inseparable over K lies in P. Its exponent lattice, generated by $(4,0),(2,2)$, has determinant 8, so $[P:K] = 216/8 = 27$.

Exercise 52.5: Ten statements about totally inseparable extensions

In this section Fraleigh reserves "totally inseparable" for a *proper* extension with one embedding. The more common "purely inseparable" convention also permits the trivial extension.

(a) False. $\mathbb{Z}_3(t)(\sqrt{t})/\mathbb{Z}_3(t)$ is a proper separable extension, since $X^2 - t$ is irreducible and has nonzero derivative.

(b) True. A totally inseparable algebraic element has some p^t-power in F.

(c) False. $\mathbb{Z}_5(y)/\mathbb{Z}_5(y^5)$ has degree 5 and only one embedding; its minimal polynomial is $X^5 - y^5$.

(d) False. $\mathbb{Z}_5(y)/\mathbb{Z}_5(y^{10})$ has inseparable degree 5 and separable degree 2, since $X^{10} - y^{10}$ has two distinct roots, each repeated five times.

(e) False. The same mixed extension has two embeddings, so is not totally inseparable either.

(f) False. The element y over $\mathbb{Z}_5(y^{10})$ in (d) is neither separable nor totally inseparable.

(g) True. The separable elements form a subfield by Exercise 51.13.

(h) False with the book's strict convention: if E/F is separable, its separable closure is E, and E/E is not a proper extension. The corrected claim allows E to equal that closure, or uses the inclusive "purely inseparable" convention.

(i) True. If E/F is not separable, its separable closure is proper, and every element outside it is totally inseparable over it.

(j) True, by the minimal-polynomial characterization of total inseparability.

Exercise 52.6: Purely inseparable elements form a field

Include the elements of F as requested. For a, b in the proposed set, choose a common power $q = p^m$ with $a^q, b^q \in F$. Then $(a+b)^q = a^q + b^q \in F$, $(ab)^q = a^q b^q \in F$, and if $b \neq 0$, $(a/b)^q = a^q/b^q \in F$. Additive inverses work in the same way. Thus it is closed under field operations and contains 0,1. Every element outside F with such a power is totally inseparable, by the power criterion, so these closure calculations prove the assertion. In characteristic zero no proper totally inseparable elements exist, and the set is simply F.

Exercise 52.7: Perfection and surjective Frobenius

If F is perfect and $a \in F$, choose a root b of $X^p - a$ in a closure. That polynomial equals $(X-b)^p$, so b has only one conjugate. Separability forces its degree to be 1; hence $b \in F$ and $a = b^p$. Thus $F^p = F$. Conversely suppose $F^p = F$. An inseparable irreducible polynomial would have derivative zero, so have form $\sum a_j X^{pj}$. Write $a_j = b_j^p$. Then it equals $(\sum b_j X^j)^p$, contradicting irreducibility. Therefore all irreducibles are separable, and F is perfect.

Exercise 52.8: Perfection through a finite extension

Let $n = [E : F]$. Frobenius is an isomorphism from E onto E^p carrying F onto F^p, so $[E^p : F^p] = n$. If $F^p = F$, then $F \subseteq E^p \subseteq E$ and both $[E^p : F]$ and $[E : F]$ equal n, forcing $E^p = E$. Conversely, if $E^p = E$, the equality above reads $[E : F^p] = n$. In the tower $F^p \subseteq F \subseteq E$, it follows that $n = n[F : F^p]$, so $[F : F^p] = 1$ and $F^p = F$. Finiteness of n is essential for this degree argument.